

Thème 3 — Fractions rationnelles • Corrigé Facile

Corrigé 1

$$\frac{2}{(x-1)(x+3)} = \frac{A}{x-1} + \frac{B}{x+3}. \text{ Astuce « multiplier-évaluer » : } A = \frac{2}{x+3} \Big|_{x=-1} = \frac{2}{4} = \frac{1}{2},$$

$$B = \frac{2}{x-1} \Big|_{x=-3} = \frac{2}{-4} = -\frac{1}{2}. \text{ Contrôle : } \frac{1}{2}(x+3) - \frac{1}{2}(x-1) = \frac{1}{2} \cdot 4 = 2 \checkmark. \text{ Donc}$$

$$\int \frac{2}{(x-1)(x+3)} dx = \frac{1}{2} \ln|x-1| - \frac{1}{2} \ln|x+3| + C.$$

Corrigé 2

$$\frac{1}{(x-2)(x-4)} = \frac{A}{x-2} + \frac{B}{x-4}. A = \frac{1}{x-4} \Big|_{x=2} = \frac{1}{-2} = -\frac{1}{2}, B = \frac{1}{x-2} \Big|_{x=4} = \frac{1}{2}. \text{ Contrôle :}$$

$$-\frac{1}{2}(x-4) + \frac{1}{2}(x-2) = -\frac{1}{2}x + 2 + \frac{1}{2}x - 1 = 1 \checkmark. \text{ Donc}$$

$$\int \frac{1}{(x-2)(x-4)} dx = -\frac{1}{2} \ln|x-2| + \frac{1}{2} \ln|x-4| + C.$$

Corrigé 3

$$\frac{3}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2}. A = \frac{3}{x-2} \Big|_{x=-1} = \frac{3}{-3} = -1, B = \frac{3}{x+1} \Big|_{x=2} = \frac{3}{3} = 1.$$

$$\text{Contrôle : } -1(x-2) + 1(x+1) = -x + 2 + x + 1 = 3 \checkmark. \text{ Donc}$$

$$\int \frac{3}{(x+1)(x-2)} dx = -\ln|x+1| + \ln|x-2| + C.$$

Corrigé 4

$$\frac{5}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}. A = \frac{5}{x+2} \Big|_{x=3} = \frac{5}{5} = 1, B = \frac{5}{x-3} \Big|_{x=-2} = \frac{5}{-5} = -1.$$

$$\text{Contrôle : } 1(x+2) - 1(x-3) = 5 \checkmark. \text{ Donc}$$

$$\int \frac{5}{(x-3)(x+2)} dx = \ln|x-3| - \ln|x+2| + C.$$

Corrigé 5

$$\frac{4}{x(x-4)} = \frac{A}{x} + \frac{B}{x-4}. A = \frac{4}{x-4} \Big|_{x=0} = \frac{4}{-4} = -1, B = \frac{4}{x} \Big|_{x=4} = \frac{4}{4} = 1. \text{ Contrôle :}$$

$$-1(x-4) + 1 \cdot x = 4 \checkmark. \text{ Donc}$$

$$\int \frac{4}{x(x-4)} dx = -\ln|x| + \ln|x-4| + C.$$